MIL-P-46421B (PA)
10 July 1975
SUPERSEDING
MIL-P-46421A (MU)
30 April 1964

MILITARY SPECIFICATION
PRIMER, PERCUSSION, M57
METAL PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M57.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8838012 - Cup, Battery
8838015 - Plug, Firing
8838016 - Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwgs.) 8838012, 8838015 and 8838016, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.3).

3.4 Strength of brazed joint.-The brazed joint securing the head and body shall withstand a pull of 5500 pounds without evidence of separation, when tested as specified in 4.4.2.

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent.

3.6 First article inspection.-This specification makes provision for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Initial production inspection.

4.2.1 At the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of (25) twenty-five complete set of parts (i.e.,(25) twenty-five each of every component) which have been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations

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or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cup, battery (see dwg. 8838012).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Outside diameter, min.	Gage
102.	Thickness through bottom, maximum (max)	Gage
103.	Length	Gage
Minor:		
201.	Diameter of hole	Gage
202.	Thickness through bottom, min.	Gage
203.	Inside diameter	Gage
204.	Outside diameter, max	Gage
205.	Flat missing from surface at open end	Visual
206.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.2 Plug, firing (see dwg. 8838015).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Large diameter, min.	Gage
102.	Length and profile of point	Gage
103.	Length of pilot, min.	Gage
104.	Total length	Gage
Minor:		
201.	Large diameter, max.	Gage
202.	Length of pilot, max.	Gage
203.	Diameter of pilot	Gage
204.	Finish improper	Visual
205.	Evidence of poor workmanship (see 3.5)	Visual

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4.3.2.3 Body (see dwg. 8838017 covering a detail of dwg. 8838016).

Categories	Defects	Method of Inspection
Critical:		
1.	Flash hole missing	Gage
2.	Crack or split	Visual
Major:		
101.	Outside diameter	Gage
102.	Wall thickness, min	Gage
103.	Location of first two pairs of holes	Gage
104.	Diameter of holes	Gage
105.	Required plug gage fails to pass through body	Gage
Minor:		
201.	Wall thickness, max	Gage
202.	Total length	Gage
203.	Finish improper	Visual
204.	Evidence of poor workmanship (see 3.5)	Visual

4.3.2.4 Head (see dwg. 8838018 covering a detail of dwg. 8838016).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Total length	Gage
202.	Depth of body cavity	Gage
203.	Chamfer at opening of body cavity missing or inadequate	Visual
204.	Radius at bottom of body cavity missing or inadequate	Visual
205.	Evidence of poor workmanship (see 3.5)	Visual

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4.3.2.5 Metal parts assembly (see dwg 8838016).

Categories	Defects	Method of Inspection
Critical:		
1.	Head assembled to wrong end of primer body	Visual
Major:		
101.	Total length, max	Gage
102.	Diameter of body, max	Gage
103.	Concentricity between diameter under flange and flange	Gage
104.	Diameter under flange	Gage
105.	Diameter of battery cup cavity	Gage
106.	Depth to bottom of percussion element cavity	Gage
107.	Diameter of percussion element cavity	Gage
108.	Thickness of flange, max	Gage
109.	Crack or split	Visual
110.	Flash hole in head missing or obstructed	Visual
111.	Braze incomplete	Visual
112.	Protective finish of body missing or damaged to the extent of exposing base metal	Visual
Minor:		
201.	Diameter of flange	Gage
202.	Depth of battery cup cavity	Gage
203.	Concentricity between percussion element cavity and flash hole	Gage
204.	Diameter of chamfer at mouth of percussion element cavity	Gage
205.	Diameter of flash hole	Gage
206.	Diameter of chamfer at mouth of flash hole	Gage
207.	Thickness of flange, min	Gage CD
208.	Radii missing	Visual
209.	Chamfer(s) missing or inadequate	Visual
210.	Finish improper	Visual
211.	Evidence of poor workmanship (see 3.5)	Visual

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4.3.2.6 Diaphragm (see dwg. 7548518).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:	AQL 1.50 percent	
201.	Large outside diameter	Gage
202.	Small outside diameter	Gage
203.	Total length	Gage

4.3.3 Testing

4.3.3.1 Hydrostatic test (see 3.3) - Critical defect.- This test shall be conducted 100 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom.

4.3.3.2 Strength of brazed joint (see 3.4) - Critical defect.-A sample of 50 assemblies shall be selected from each lot for this test. If one or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2 using equipment as specified in 4.3.4.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and test prescribed in this section is identified under the "Method of Inspection" heading in 4.3.2 and 4.3.3 (see 6.2).

4.4 Test methods and procedures.

4.4.1 Hydrostatic test.-Prior to machining, the tubing for primer fabrication shall be assembled into the test equipment specified in 4.3.4 and shall be tested for not less than five seconds at not less than the hydrostatic pressure calculated from the following formula:

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P = (2ST times .9) divided by D

where:

P = test pressure in pounds per square inch (psi)
S = min. specified yield strength, psi
T = specified min. wall thickness in inches
D = specified max. outside diameter in inches

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

4.4.2 Strength of brazed joint.-The metal parts assembly shall be tested in a universal testing machine by applying the specified load in the axial direction. The rate of travel of the testing machine shall be 0.100 to 0.125 inch per minute. The assembly shall be observed for compliance with the applicable requirement. The sample assemblies subjected to this test shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-Preservation and packaging of parts for primer, percussion, M57, shall be in accordance with Military Standard MIL-STD-1169.

5.2 Marking.-Marking shall be in accordance with Military Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data card shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. First article inspection provision.

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6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in Section 4, or on the Equipment Lists referenced on the Equipment Tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which specified and may include commercial equipment which the contractor proposes to use. Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification

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number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1390-A092

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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